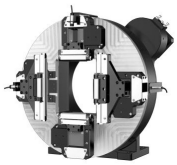
SF6024T · TUBE LASER CUTTING MACHINE (FOLLOW-UP SUPPORT) · 1.5 – 6 KW

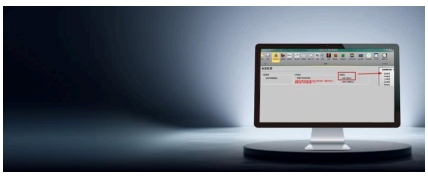
SF6024T
TECHNICAL SPECIFICATION

TUBE LASER CUTTING MACHINE (FOLLOW-UP SUPPORT)









Ø0.8 – 9.1"	6.6 – 21.3 ft	328 ft/min	1 G
ROUND TUBE	TUBE LENGTH	RAPID TRAVERSE	ACCELERATION

FULL TECHNICAL SPECIFICATION · SF6024T · 1.5 – 6 KW

CONTENTS

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- CSA-approved electrical panel
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SOLD & SERVICED IN CANADA

by NovoTech Machine Tools — Mississauga, ON

01 · MODEL · TUBE LASER CUTTING MACHINE (FOLLOW-UP SUPPORT)

SF6024T

Versatile & efficient





Senfeng's campus in Jinan, China

1 SUPERIOR CUTTING STABILITY

2 POWERFUL DYNAMIC PERFORMANCE

3 AUTOMATED SYSTEM FOR GREATER EFFICIENCY

4 PNEUMATIC SELF-CENTERING CHUCK

**The picture is for reference; the actual appearance and size shall prevail.*

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02 · TECHNICAL PARAMETERS

TECHNICAL PARAMETERS · SF6024T

NO.	ITEM	PARAMETER	REMARKS
1	**Tube length range**	6.6 – 21.3 ft (2,000 – 6,500 mm)	
2	**Tube diameter range**	Ø0.8 – 9.1″ (Ø20 – 230 mm)	National standard tube
3	**Side length of square tube**	□0.8 – 9.1″ (20 – 230 mm)	National standard tube
4	**X / Y axis positioning accuracy**	±0.002″ (±0.05 mm)	
5	**X / Y axis repeated positioning accuracy**	±0.001″ (±0.03 mm)	
6	**Maximum rapid traverse speed**	328 ft/min (100 m/min)	As tube weight increases, processing speed and moving parameters decrease.
7	**Maximum chuck rotation speed**	100 rpm	
8	**Maximum acceleration**	1.0 G	
9	**Weight**	6,945 lb (3,150 kg)	Depends on actual factory delivery
10	**Maximum load**	573 lb (260 kg)	Tube weight ≤28.9 lb/ft (43 kg/m)
11	**Dimensions (L × W × H)**	31.3 × 6.4 × 7.2 ft (9,530 × 1,950 × 2,200 mm)	Depends on actual factory delivery
12	**Phase**	Three-phase	
13	**Rated supply voltage**	380 V — in Canada: 208 / 240 / 480 / 600 V 3-phase 60 Hz through the transformer NovoTech supplies	
14	**Frequency**	50 Hz	
15	**Main power protection level**	IP54	

1. The achievable workpiece accuracy partly depends on factors such as the workpiece type, pretreatment condition, size and position.

2. The technical parameters above are subject to change without prior notice. The final technical parameters shall be as specified in the actual order agreement.

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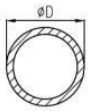
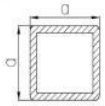
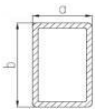
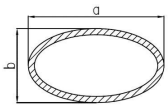
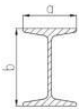
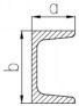
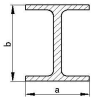
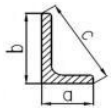
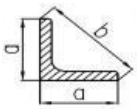
03 · TECHNICAL PARAMETERS

PRECAUTIONS · SF6024T

1. Pipes being processed should be free from significant rust, as it can affect the cutting section quality.
2. Pipes should not be stored in the open air and should be placed flat, preferably with a surface oil coating or packaging. Ensure that the front end is even and the burr height at both ends is ≤0.20" (5 mm). Bundled pipes should have consistent lengths, with a deviation of less than 11.8" (300 mm).
3. For welded pipes, the external weld seam should be relatively flat, with a height of less than 0.012" (0.3 mm). The internal weld seam should have a height of less than 0.08" (2 mm).
4. The distortion and bending of the pipe should be lower than the tolerances specified in "GB/T 6728-2002 Dimensions, Shape, Weight, and Permissible Deviations of Cold-formed Hollow Sections for Structural Use."
5. The pipe should be straight, with a bending curvature of less than 0.012" per ft (1 mm/1 m; 3 mm over 6 m).
6. For the lengthwise torsion of the pipe, the total torsion should be less than 0.02 % of the total length.
7. The outer diameter tolerance of the profile should not exceed ±0.5 % of the outer diameter, with a minimum of 0.008" (0.2 mm)(according to the standard deviation level D4 specified in GB/T 17395-1998 for standardized outer diameter deviations).
8. When loading the pipes, ensure the safety of personnel and equipment. Bundle loading should be carried out using cranes operated by qualified and certified professionals. The crane should have both fast and slow speed functions, and slow speed should be used when approaching the loading platform.
9. Considering material errors, the positional error should be within IT12, and the shape and profile error should be within IT12.
10. The surface roughness of the pipe cross-section varies with the material and thickness.
11. Users must perform regular maintenance of the machine as instructed in the manual. Failure to do so may result in damage to the machine, and the manufacturer reserves the right to refuse repairs in such cases.
12. To ensure cutting effectiveness and minimize pipe deformation, the pipe wall thickness should meet the following criteria based on the pipe diameter:
    1. For pipe diameters greater than Ø2" (Ø50 mm): 1/40 ≤ wall thickness ≤ 1/10 of the pipe diameter.
    2. For pipe diameters smaller than Ø2" (Ø50 mm): 0.031" (0.8 mm) ≤ wall thickness ≤ 1/10 of the pipe diameter.
    3. Where the bending deflection exceeds the relevant standards, the cutting quality cannot be guaranteed.
13. Equipment temperature requirements: 32 – 104 °F (0 – 40 °C)(if the ambient temperature at the installation site exceeds this range, the buyer shall give written notice to the supplier at the time of purchase); the laser source should be placed in an environment of 64 – 77 °F (18 – 25 °C). If the ambient temperature at the installation site exceeds this range, air conditioning is recommended.
14. Equipment humidity requirements: ≤70 % (laser source only).
15. Cutting gas purity requirements: nitrogen ≥99.7 %; oxygen ≥99.5 %.

04 · CUTTING PARAMETERS

CLAMPING RANGE · SF6024T

TYPE	CROSS-SECTION	CLAMPING RANGE
Round tube		Ø0.4 – 9.1″ (Ø10 – 230 mm) Ø10 – Ø230 mm
Square tube		a: 0.4 – 9.1″ (10 – 230 mm) □10 × 10 – □230 × 230 mm
Rectangular tube		a: 0.4 – 9.1″ (10 – 230 mm) · b: 0.4 – 9.1″ (10 – 230 mm) □10 × 10 – □230 × 230 mm
Oval tube		For a better clamping effect, please provide a detailed drawing so the jaws can be designed accordingly.
I-beam		min. a: 0.4″ (10 mm) · max. b: 7.9″ (200 mm) Standard configuration: GB No. 10 – 20 · optional accessory: No. 22
U-beam (channel)		min. a: 0.4″ (10 mm) · max. b: 7.1″ (180 mm) Standard configuration: GB No. 5 – 20 · optional accessory: No. 22
H-beam		a: 0.4 – 7.9″ (10 – 200 mm) · b: 0.4 – 7.9″ (10 – 200 mm)
Equal angle steel		a: 0.4 – 4.3″ (10 – 110 mm) · max. b: 5.5″ (140 mm) GB No. 2 – 11
Unequal angle steel		a: 0.4 – 3.0″ (10 – 75 mm) · b: 0.4 – 5.9″ (10 – 150 mm) · max. c: 6.1″ (155 mm) GB No. 2.5/1.6 – 14/9

To ensure better machining accuracy, the tube diameter must be ≥0.8″ (20 mm).

05 · CUTTING PARAMETERS · SF6024T

CARBON STEEL

Cutting speed by laser source power in inches per minute, with Senfeng's metres per minute and the assist gas below each value.

Material	Wall thickness	1.5 kW	2 kW	3 kW	4 kW	6 kW
CARBON STEEL	**0.039″** 1 mm	394 – 512 10 – 13 · N_2 / air	591 – 787 15 – 20 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	1,181 – 1,772 30 – 45 · N_2 / air
	0.079″ 2 mm	157 – 197 4 – 5 · O_2	236 – 315 6 – 8 · N_2 / air	512 – 669 13 – 17 · N_2 / air	591 – 709 15 – 18 · N_2 / air	787 – 984 20 – 25 · N_2 / air
	0.118″ 3 mm	118 – 157 3 – 4 · O_2	118 – 157 3 – 4 · O_2	276 – 394 7 – 10 · N_2 / air	315 – 394 8 – 10 · N_2 / air	472 – 551 12 – 14 · N_2 / air
	0.157″ 4 mm	91 – 110 2.3 – 2.8 · O_2	98 – 118 2.5 – 3 · O_2	197 – 315 5 – 8 · N_2 / air	236 – 315 6 – 8 · N_2 / air	276 – 315 7 – 8 · N_2 / air
	0.197″ 5 mm	71 – 91 1.8 – 2.3 · O_2	87 – 102 2.2 – 2.6 · O_2	118 – 126 3 – 3.2 · O_2	118 – 126 3 – 3.2 · O_2	197 – 236 5 – 6 · N_2 / air
	0.236″ 6 mm	63 – 79 1.6 – 2.0 · O_2	71 – 79 1.8 – 2 · O_2	91 – 98 2.3 – 2.5 · O_2	98 – 110 2.5 – 2.8 · O_2	177 – 197 4.5 – 5 · N_2 / air
	0.315″ 8 mm	/	47 – 63 1.2 – 1.6 · O_2	71 – 87 1.8 – 2.2 · O_2	79 – 91 2 – 2.3 · O_2	87 – 98 2.2 – 2.5 · O_2
	0.394″ 10 mm	/	/	55 – 71 1.4 – 1.8 · O_2	63 – 79 1.6 – 2 · O_2	79 – 91 2.0 – 2.3 · O_2
	0.472″ 12 mm	/	/	39 – 55 1 – 1.4 · O_2	39 – 55 1 – 1.4 · O_2	75 – 83 1.9 – 2.1 · O_2
	0.551″ 14 mm	/	/	/	/	55 – 67 1.4 – 1.7 · O_2
	0.630″ 16 mm	/	/	/	/	47 – 55 1.2 – 1.4 · O_2
	0.709″ 18 mm	/	/	/	/	31 0.8 · O_2
	0.787″ 20 mm	/	/	/	/	/

1. When processing carbon steel tubes with a wall thickness over 0.197″ (5 mm) and a diameter or side length under 2″ (50 mm), oxygen is not recommended as the assist gas. For tubes of these dimensions, please consult the process engineering team.

2. The parameters are for reference only. The values above are theoretical; tube dimensions, equipment speed and other factors will affect the cutting speed.

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06 · CUTTING PARAMETERS · SF6024T

STAINLESS STEEL · ALUMINUM ALLOY

Cutting speed by laser source power in inches per minute, with Senfeng's metres per minute and the assist gas below each value.

Material	Wall thickness	1.5 kW	2 kW	3 kW	4 kW	6 kW
STAINLESS STEEL	0.039″ 1 mm	591 – 787 15 – 20 · N_2 / air	591 – 787 15 – 20 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	1,181 – 1,772 30 – 45 · N_2 / air
	0.079″ 2 mm	276 – 394 7 – 10 · N_2 / air	315 – 551 8 – 14 · N_2 / air	512 – 669 13 – 17 · N_2 / air	591 – 709 15 – 18 · N_2 / air	787 – 984 20 – 25 · N_2 / air
	0.118″ 3 mm	177 – 217 4.5 – 5.5 · N_2 / air	236 – 276 6 – 7 · N_2 / air	236 – 394 6 – 10 · N_2 / air	315 – 472 8 – 12 · N_2 / air	472 – 709 12 – 18 · N_2 / air
	0.157″ 4 mm	79 – 98 2.0 – 2.5 · N_2 / air	110 – 138 2.8 – 3.5 · N_2 / air	197 – 236 5 – 6 · N_2 / air	236 – 276 6 – 7 · N_2 / air	394 – 472 10 – 12 · N_2 / air
	0.197″ 5 mm	55 – 71 1.4 – 1.8 · N_2 / air	59 – 98 1.5 – 2.5 · N_2 / air	118 – 157 3 – 4 · N_2 / air	157 – 177 4 – 4.5 · N_2 / air	276 – 315 7 – 8 · N_2 / air
	0.236″ 6 mm	28 – 35 0.7 – 0.9 · N_2 / air	39 – 59 1 – 1.5 · N_2 / air	91 – 118 2.3 – 3 · N_2 / air	118 – 138 3 – 3.5 · N_2 / air	177 – 197 4.5 – 5 · N_2 / air
	0.315″ 8 mm	/	/	39 – 59 1.0 – 1.5 · N_2 / air	59 – 71 1.5 – 1.8 · N_2 / air	98 – 138 2.5 – 3.5 · N_2 / air
	0.394″ 10 mm	/	/	31 – 39 0.8 – 1 · N_2 / air	39 – 47 1 – 1.2 · N_2 / air	47 – 79 1.2 – 2 · N_2 / air
	0.472″ 12 mm	/	/	/	/	39 – 47 1 – 1.2 · N_2 / air
	0.551″ 14 mm	/	/	/	/	31 – 39 0.8 – 1 · N_2 / air
	0.630″ 16 mm	/	/	/	/	20 – 24 0.5 – 0.6 · N_2 / air
ALUMINUM ALLOY	0.039″ 1 mm	394 – 512 10 – 13 · N_2 / air	591 – 787 15 – 20 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	787 – 984 20 – 25 · N_2 / air	1,181 – 1,772 30 – 45 · N_2 / air
	0.079″ 2 mm	157 – 197 4 – 5 · N_2 / air	315 – 512 8 – 13 · N_2 / air	512 – 669 13 – 17 · N_2 / air	984 – 1,378 25 – 35 · N_2 / air	787 – 984 20 – 25 · N_2 / air
	0.118″ 3 mm	59 – 98 1.5 – 2.5 · N_2 / air	157 – 177 4 – 4.5 · N_2 / air	236 – 315 6 – 8 · N_2 / air	591 – 709 15 – 18 · N_2 / air	551 – 630 14 – 16 · N_2 / air
	0.157″ 4 mm	39 – 51 1 – 1.3 · N_2 / air	98 – 118 2.5 – 3 · N_2 / air	157 – 197 4 – 5 · N_2 / air	315 – 472 8 – 12 · N_2 / air	315 – 394 8 – 10 · N_2 / air
	0.197″ 5 mm	/	59 – 79 1.5 – 2 · N_2 / air	98 – 138 2.5 – 3.5 · N_2 / air	236 – 276 6 – 7 · N_2 / air	197 – 236 5 – 6 · N_2 / air
	0.236″ 6 mm	/	/	79 – 91 2 – 2.3 · N_2 / air	157 – 177 4 – 4.5 · N_2 / air	138 – 157 3.5 – 4 · N_2 / air
	0.315″ 8 mm	/	/	31 – 51 0.8 – 1.3 · N_2 / air	98 – 118 2.5 – 3 · N_2 / air	59 – 79 1.5 – 2 · N_2 / air
	0.394″ 10 mm	/	/	/	39 – 51 1 – 1.3 · N_2 / air	39 – 47 1 – 1.2 · N_2 / air
	0.472″ 12 mm	/	/	/	/	24 – 28 0.6 – 0.7 · N_2 / air
	0.551″ 14 mm	/	/	/	/	16 – 24 0.4 – 0.6 · N_2 / air

1. When processing carbon steel tubes with a wall thickness over 0.197″ (5 mm) and a diameter or side length under 2″ (50 mm), oxygen is not recommended as the assist gas. For tubes of these dimensions, please consult the process engineering team.

2. The parameters are for reference only. The values above are theoretical; tube dimensions, equipment speed and other factors will affect the cutting speed.

07 · COST-BENEFIT ANALYSIS

COST-BENEFIT ANALYSIS · SF6024T

If the cutting auxiliary gas is dried compressed air, the cost includes the air compressor's electricity, the machine's power consumption and the consumables (protective lenses, cutting nozzles).

Item		3 kW			4 kW			6 kW		
		Air	O_2	N_2	Air	O_2	N_2	Air	O_2	N_2
PEAK POWER CONSUMPTION	Laser source (kW)	10			16			16		
	Chiller (kW)	3			6			6		
	Air compressor (kW)	15	/	/	15	/	/	15	/	/
	Host (kW)	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5	12.5
	Dust removal (kW)	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
	Loading & unloading (kW)	3.85	3.85	3.85	3.85	3.85	3.85	3.85	3.85	3.85
Consumables and gas consumption (CNY/h)		0.5	4.5	60.5	0.5	4.5	60.5	0.5	4.5	60.5
Total power (kW)		45.85	30.85	30.85	54.85	39.85	39.85	54.85	39.85	39.85
Total power consumption (kWh per hour)		27.51	18.51	18.51	32.91	23.91	23.91	32.91	23.91	23.91
Total operating cost (RMB per hour, at 1 RMB/kWh)		28.01	23.01	79.01	33.41	28.41	84.41	33.41	28.41	84.41

1. The electricity and gas prices are Senfeng's reference values in China (RMB) and may vary by region.

2. Auxiliary gas consumption varies with plate thickness. The oxygen data is based on cutting 0.315″ (8 mm) carbon steel and nitrogen on 0.039″ (1 mm) stainless steel; for reference only, actual usage may vary.

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08 · CONFIGURATION LIST

CONFIGURATION LIST · SF6024T

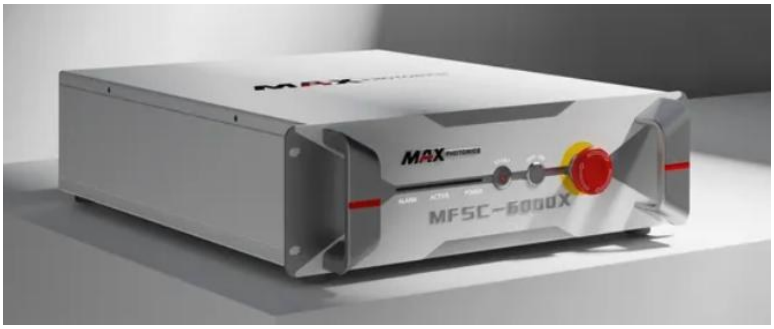
NO.	ITEM	QTY	BRAND
		LASER SOURCE	
1	Fiber laser source	1	Maxphotonics
		LASER CUTTING HEAD	
1	Laser cutting head	1	Raytools
		MACHINE TOOL · HOST	
1	Transmission system	8	SENFENG
2	Machine tool and accessories	1	SENFENG
3	Reducer	6	TECHMECH / MOTOREDUCER
4	Cutting gas circuit system	1	AVENTICS (Germany)
5	Main machine air circuit system	1	AIRTAC (Taiwan, China)
6	AC servo motor and driver	8	SCHNEIDER (France) / INOVANCE
7	Water chiller	1	HANLI
		CNC CUTTING SOFTWARE SYSTEM	
1	CNC system	1	FSCUT 3000DE-G
2	Nesting software	1	TubesT professional nesting software
		AUTOMATION SYSTEM (OPTIONAL)	
1	Loading system	1	Fully automatic / simplified fully automatic / semi-automatic (three options available)
2	Unloading system	1	Standard: 6.6 ft (2 m) · optional: 10 ft (3 m) / 13 ft (4 m)

1. This is Senfeng's optimized configuration. Changes in brand or configuration may cause irreversible effects.

2. The warranty period for the entire machine (excluding consumables, non-force majeure natural disasters, war, improper operation, and human damage) is 1 year.

09 · CORE COMPONENTS

SF6024T LASER SOURCE



MAX fiber laser source

LASER SOURCE

1. **Small fiber core:** better beam quality.
2. **Faster cutting speed:** enhanced cutting efficiency.
3. **Improved high-reflection resistance:** new optical components with a multi-layer anti-reflection design.
4. **Enhanced intelligence:** real-time dynamic monitoring of operating indicators.
5. **High electrical-optical conversion:** over 40 % efficiency, reducing power consumption.
6. **Upgraded components:** new-generation pump sources offer greater reliability and stability, with improved beam quality and 30 – 40 % more brightness thanks to patented nonlinear suppression technology.
7. **Lightweight and compact:** small size and compact design for easier integration.
8. **Enhanced safety:** improved safety features.

LASER SOURCE — MEDIUM AND THIN PLATE

1. **Small fiber core:** better beam quality.
2. More efficient cutting of medium and thin plates.
3. **Improved high-reflection resistance:** new optical components with a multi-layer anti-reflection design.
4. **Enhanced intelligence:** real-time dynamic monitoring of operating indicators.
5. **High electrical-optical conversion:** over 40 % efficiency, reducing power consumption.
6. **Versatile applications:** the modulation frequency range is upgraded to 1 – 20 kHz, suitable for a wide variety of processes.
7. **Lightweight and compact:** small size and compact design for easier integration.
8. **High sealing performance:** sealing and reliability further improved, with the protection rating upgraded to IP66.

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09 · CORE COMPONENTS

SF6024T LASER HEAD



3 kW



6 kW

1. **Optimized design:** an enhanced optical configuration and streamlined airflow design significantly improve cutting quality and efficiency.
2. **Auto-focusing:** minimizes human intervention, with a focusing accuracy of 0.002″ (0.05 mm).
3. **Multiple protective lenses:** the collimation module includes additional protective lenses that safeguard the collimating lens.
4. **Drawer-style lens holder:** quick and easy replacement of the protective lenses.
5. **High compatibility:** equipped with QBH and QD fiber interfaces.

OPTIONAL BOCI CUTTING HEAD



BOCI cutting head (optional)

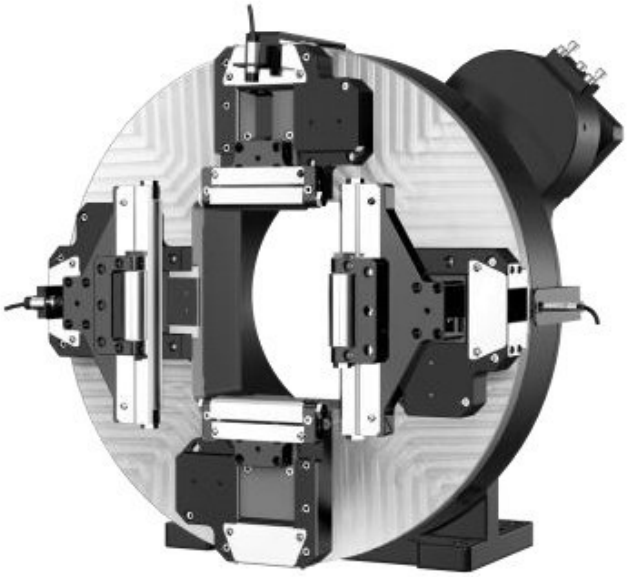
1. Closed-loop system with active feedback alarms for precise issue detection.
2. Protective lens temperature monitoring detects contamination during operation and triggers immediate alarms to stop the machine, preventing damage such as head burning or lens breakage.
3. Monitors cutting gas pressure in real time, triggering alarms for insufficient gas supply, low pressure or unstable pressure.
4. The anti-collision design of the cutting head enables easy replacement of components after an impact, reducing downtime.

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10 · CORE COMPONENTS

SF6024T CHUCK





1. **Pneumatic self-centering chuck:** fast clamping speed and high efficiency.
2. **Low-inertia chuck:** made of aerospace-grade aluminum — lightweight, low inertia, high rotation speed, lower power consumption and lower operating costs.
3. **Fully enclosed chuck protection:** keeps dust and debris out, ensuring smooth operation without stoppages.
4. **Internal gears and external jaws:** better dust resistance, longer lifespan, high precision, and easier adjustment and maintenance.
5. **Integrated chuck structure:** superior dust protection and greater machine stability.
6. **Square-opening front chuck:** expands the clamping range and broadens industry applications.

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11 · HOST MACHINE

SF6024T HOST MACHINE





1 · Machine bed

STABILITY, VIBRATION REDUCTION AND DURABILITY

1. The machine bed is welded from rectangular tubes, giving greater stability, a higher load-bearing capacity, significantly better vibration damping and better operational performance.
2. The design is based on structural dynamic principles and variable analysis with the finite element method (FEM). After precise welding, the bed undergoes stress-relief annealing and secondary aging treatment, followed by precision machining on a large gantry milling machine — high rigidity and a long service life.

2 · Follower support

EFFECTIVELY SUPPORTS TUBES

1. The segmented special reducing-wheel auxiliary support structure adapts to pipes of different diameters and shapes.
2. It effectively addresses the reduced processing accuracy caused by the sagging of long pipes.





3 · Exceptional tube processing capability

WIDE SIZE RANGE & HIGH SINGLE-TUBE LOAD CAPACITY

1. **Size range:** round tubes of Ø0.8 – 9.1″ (Ø20 – 230 mm); square tubes of 0.8 × 0.8 – 9.1 × 9.1″ (20 × 20 – 230 × 230 mm).
2. **Supported shapes:** round, square and rectangular tubes.
3. **High single-tube load:** each tube can weigh up to 441 lb (200 kg), outstanding value in tube cutting.

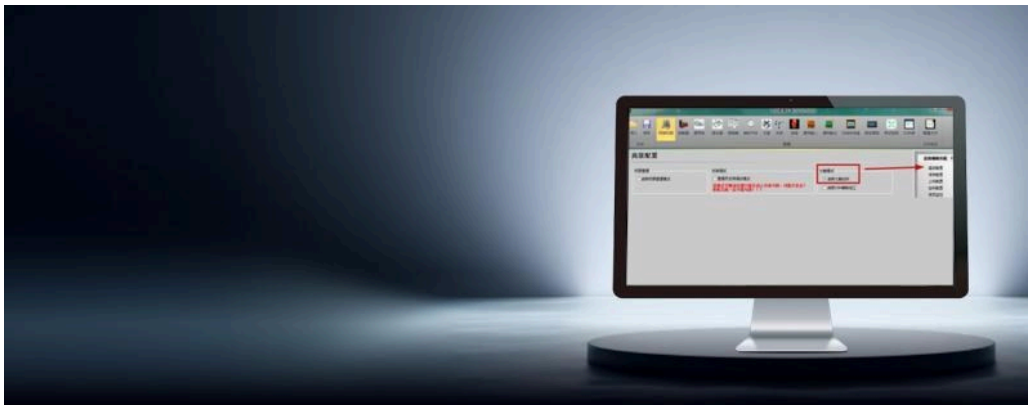
4 · Powerful dynamic performance

ULTRA-HIGH PRECISION

The all-new tube laser cutting machine combines exceptional performance with ultra-high precision, fully meeting the stringent requirements of the tube cutting industry.

12 · CNC SYSTEM

FSCUT3000DE-G CNC SYSTEM



Quick operation & efficient cutting

FSCUT3000DE-G is a bus control system specifically developed for tube cutting. It supports high-precision, high-efficiency cutting of square tubes, round tubes, racetrack-shaped and oval tubes, as well as angle steel and channel steel. Its performance features are as follows:

- Real-time tube center deviation compensation reduces clamping requirements and improves efficiency, ensuring higher piercing accuracy and more precise cutting.
- Built on a real-time bus control system with integrated following control, it delivers faster corner cutting, smoother corner tracking and more reliable cutting quality.
- Supports the follow-up support function, ensuring consistent cutting quality for full-length tubes.
- Compatible with hollow chuck feeding and cutting, significantly reducing the overall machine length and enhancing automation.
- Supports automatic loading / unloading and cyclic processing, including standard automated feeding actions.
- Trajectory accuracy of 0.0008" (0.02 mm), positioning accuracy of 0.00004" (0.001 mm) and repeatability of 0.0001" (0.003 mm).
- Supports real-time encoder feedback and error measurement tools to help optimize the machine's motion parameters.
- Supports a tube process library for convenient export and storage of cutting processes.
- Supports seamless piercing, for smoother operation and significantly improved piercing efficiency.
- Compatible with the TubePro cutting software and the professional version of the TubesT nesting software — it can process all types of tubes.

13 · OPTIONAL CONFIGURATION

SF6024T OPTIONAL CONFIGURATION

1 · 6.6 FT (2 M) UNLOADING UNIT optional



Dimensions	76.8 × 41.7 × 30.3″ (1,950 × 1,060 × 770 mm)
Maximum tube length	9.8 ft (3,000 mm)
Unloading weight	661 lb (300 kg)

2 · DUAL-ARM CHAIN-TYPE SEMI-AUTOMATIC LOADING UNIT optional



Minimum loading tube length	13.1 ft (4,000 mm)
Maximum loading tube length	20 ft (6,100 mm)
Tube section side length (or diameter)	0.8 × 0.8 – 8.7 × 8.7″ (20 × 20 – 220 × 220 mm); Ø0.8 – 8.7″ (Ø20 – 220 mm)
Maximum single tube weight	529 lb (240 kg)
Tube type	Round, square and rectangular tube, angle steel, channel steel, I-beams, etc.

3 · FULLY AUTOMATIC LOADING UNIT optional



Loading tube length	11.5 – 20 ft (3,500 – 6,100 mm)
Tube section side length (or diameter)	0.8 × 0.8 – 9.1 × 9.1″ (20 × 20 – 230 × 230 mm); Ø0.8 – 9.1″ (Ø20 – 230 mm)
Maximum single tube weight	573 lb (260 kg)
Warehouse weight capacity	6,614 lb (3 t)
Tube type	Round tube, square tube

4 · SIMPLIFIED FULLY AUTOMATIC LOADING UNIT optional



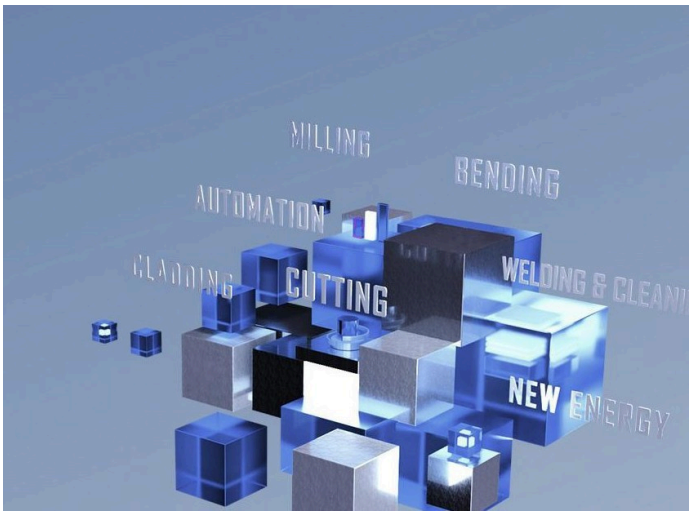
Loading tube length	11.5 – 20 ft (3,500 – 6,100 mm)
Tube section side length (or diameter)	0.8 × 0.8 – 9.1 × 9.1″ (20 × 20 – 230 × 230 mm); Ø0.8 – 9.1″ (Ø20 – 230 mm)
Maximum single tube weight	573 lb (260 kg)
Warehouse weight capacity	6,614 lb (3 t)
Tube type	Round tube, square tube

1. Equipped with an automatic sorting system that separates scrap from finished tubes without manual intervention. The receiving cart and unloading rack have an integrated design — high system integration and a compact footprint.
2. A self-developed servo-following support system automatically adjusts to the tube surface, ensuring optimal support during processing. After processing, the flip plate discharges the finished tubes, which slide into the finished product cart.

14 · AFTER-SALES SERVICE

PROVIDING GLOBAL USERS WITH AUTOMATED METAL FABRICATION SOLUTIONS

Senfeng provides automated metal fabrication solutions globally, integrating cutting, bending, welding, cladding, automation and new energy. With core technologies in laser cutting, welding and cladding, the company also offers intelligent manufacturing systems such as flexible laser processing lines and panel benders. These solutions serve industries including automotive, construction, energy and petrochemicals, and are used in over 100 countries.



03
INTERNATIONAL PCT PATENTS

577
CHINESE TECHNOLOGY PATENTS

Senfeng has developed key components such as laser generators, processing heads, the Feng Cloud system and CNC systems, used in cutting, welding, cladding and automation. The company offers a complete industry chain, including laser cutting, welding, cladding and cleaning machines, bending centers and flexible production lines. These solutions are used in sectors such as power towers, construction machinery, shipbuilding, bridges and aerospace, helping businesses accelerate production and reduce costs for greater economic benefit.

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15 · AFTER-SALES SERVICE

TRAINING · PACKAGING · INSTALLATION

TECHNICAL TRAINING TO ENHANCE CUSTOMER PRODUCTION EFFICIENCY

01
BEFORE EQUIPMENT DELIVERY
The buyer can arrange for 1 – 2 operators to attend a one-week training at the Senfeng factory. The dates are confirmed with Senfeng's customer service department.

02
DURING THE WARRANTY PERIOD
The buyer can apply for one more session of free training for 1 – 2 operators at Senfeng.

03
TRAINING
Laser principles, equipment structure, process explanation, equipment maintenance, laser safety protection, operating procedures and basic troubleshooting.

04
REQUIREMENT
Trainees must be mechanical, electrical or optical engineers and pass assessments on equipment operation, laser principles, safety and maintenance before starting work.

PACKAGING & TRANSPORTATION TO ENSURE EQUIPMENT QUALITY







01
PACKAGING
Standard packaging, suitable for long-distance transportation, moisture-proof, rust-proof and shock-resistant. Designed for full lifting, with marked lifting points and center of gravity.

02
TRANSPORTATION
Domestic transportation within China is fully handled by Senfeng, including freight and insurance.

03
PACKING LIST & CERTIFICATE
Each package includes a packing list and a certificate of conformity. The user manual and other documents are inside the box; the packing list is on the outside.

INSTALLATION PROFESSIONAL AND HIGH-QUALITY

01
INSTALLATION
NovoTech engineers, with Senfeng, install the equipment at the customer's site.

02
EQUIPMENT DEBUGGING
After installation, the equipment is debugged for the customer's use.

03
DOOR-TO-DOOR TRAINING
On-site training on equipment maintenance, safety, operating procedures and basic troubleshooting, with 7 days of guaranteed usage.

04
ON-SITE ACCEPTANCE
After the steps above, the engineer leaves the site only after the customer's acceptance (the customer can veto).

16 · AFTER-SALES SERVICE

PREMIUM CUSTOMIZATION

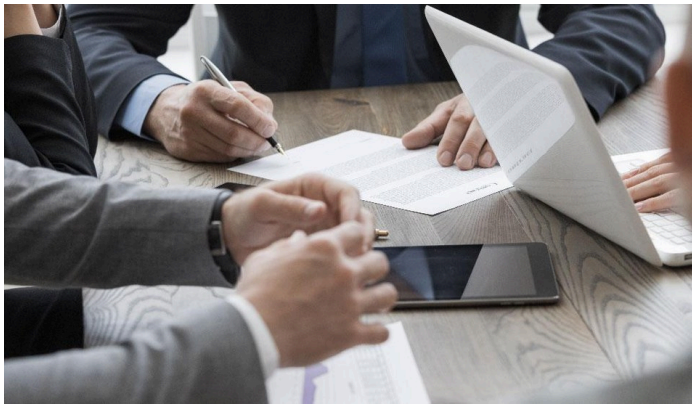
In the digital era, intelligent transformation in metal processing is essential. Building unmanned factories is key to upgrading, with customized automation solutions a top priority.



01 · Senfeng understands the client's industry status and specific production processes, identifies issues and assesses their needs.



02 · Through in-depth on-site communication, metal fabrication automation solutions are customized to the client's pain points and needs.



03 · A tailored solution, different from competitors' standardized models.



04 · A one-stop R&D and production service from design to final machine, with thorough quality checks until customer satisfaction and acceptance.

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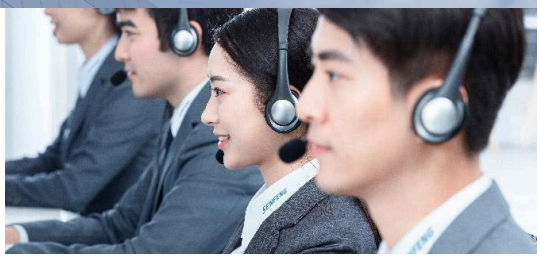
17 · AFTER-SALES SERVICE

FIVE-STAR FAST SERVICE



EFFICIENT

- Senfeng's repair hotline is available 24/7.
- A professional engineer responds to customer inquiries within 10 minutes and provides a repair plan within 1 hour.



PROFESSIONAL

- **Custom service:** service plans tailored to the customer's specific needs.
- **Service engineer certification:** each service engineer is rigorously trained and assessed before being certified to work.
- **Common issues training:** a manual of common issues for each equipment model, with certified engineers training customers.
- **Online guidance:** experienced senior engineers support customers by phone or video call.
- **Professional technical support:** the equipment is properly calibrated at the first installation, and similar issues are resolved in one go.





COMPREHENSIVE

- **Pre-service:** theoretical and hands-on operation training, common fault self-diagnosis training, quick repair guidance and usage reminders.
- **Regular service:** regular maintenance reminders, on-site services and periodic promotional activities.
- **Value-added service:** equipment software and hardware upgrades, financing lease services and extended warranty services.



18 · AFTER-SALES SERVICE

CUSTOMER TESTIMONIALS

Senfeng machines and customers around the world.



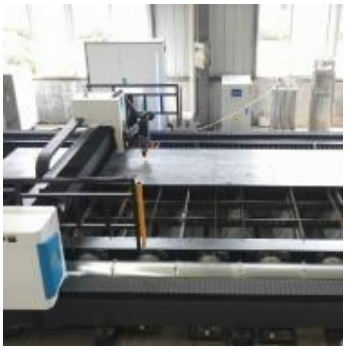
















NovoTech MACHINE TOOLS Inc.

LET'S FIND THE RIGHT TUBE LASER FOR YOUR SHOP.

Call or visit us for quotes, demonstrations, installation and service. In-stock machines ship fast from our Mississauga, ON location.

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+1 647 526 5510

WEB & EMAIL

NovoTechMachineTools.com
info@NovoTechMachineTools.com

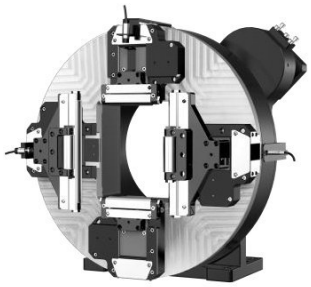
ADDRESS

1365 Mid-Way Blvd, 26
Mississauga, ON L5T 2J5
Canada

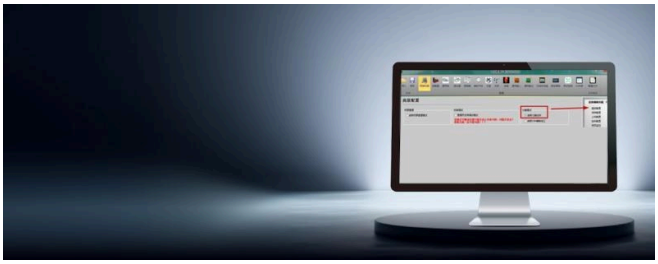
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Specifications subject to change without notice. Imperial values are converted from Senfeng's metric data and rounded: sheet thicknesses in decimal inches, cutting speeds in inches per minute. Model numbers are manufacturer designations. Photos may show optional equipment.